THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. How many lakhs make 3 crore? 1

(a) 1300 **(b)** 300 **(c)** 310 **(d)** 290

2. The value of $48 \div (6 + 2) \times 3$ is: 1

(a) 14 **(b)** 20 **(c)** 30 **(d)** 18

3. $2.5 \times 0.4 =$ 1

(a) 100 **(b)** 0.1 **(c)** 1 **(d)** 10

4. The perimeter of a square of side s is $4s$. For $s = 7$ cm it is (in cm): 1

(a) 29 **(b)** 28 **(c)** 27 **(d)** 30

5. If x° and 24° are complementary, then $x =$ 1

(a) 56° **(b)** 66° **(c)** 76° **(d)** 86°

6. A two-digit number has digits 2 and 5. The sum of the number and its reverse is: 1

(a) 78 **(b)** 76 **(c)** 77 **(d)** 79

7. Two sides of a triangle are 5 cm and 9 cm. The third side could be: 1

(a) 14 cm **(b)** 7 cm **(c)** 16 cm **(d)** 4 cm

8. How many pieces of length $\frac{1}{4}$ m can be cut from a rope of 3 m? 1

(a) $\frac{3}{4}$ **(b)** 12 **(c)** 24 **(d)** 6

9. 1,25,00,000 written in the international system has how many commas? 1

(a) 4 **(b)** 3 **(c)** 2 **(d)** 1

10. The value of $36 \div 6 + 4 \times 5 - 3$ is: 1

(a) 25 **(b)** 47 **(c)** 21 **(d)** 23

11. $4.8 \div 0.6 =$ 1

(a) 8 **(b)** 80 **(c)** 800 **(d)** 0.8

12. The rule 'multiply by 5 and add 2' for the input n is: 1

(a) $5n + 2$ **(b)** $2n + 5$ **(c)** $5(n + 2)$ **(d)** $n + 7$

13. In a linear pair one angle is 2 times the other. The smaller angle is: 1

(a) 50° **(b)** 70° **(c)** 80° **(d)** 60°

14. A two-digit number has digits 9 (tens) and 2 (units). Number minus its reverse is: 1

(a) 63 **(b)** 64 **(c)** 62 **(d)** 65

15. The angles of a triangle are x°, $2x^\circ$ and $3x^\circ$. The value of x is: 1

(a) 31 **(b)** 30 **(c)** 32 **(d)** 29

16. Scaling a drawing by $\frac{3}{4}$: a line of 20 cm becomes (cm): 1

(a) $26\frac{2}{3}$ **(b)** 30 **(c)** $7\frac{1}{2}$ **(d)** 15

17. 45,60,000 in the international system is: 1

(a) 456,000 **(b)** 4,560,000 **(c)** 45,60,000 **(d)** 45,600,000

18. Evaluate: $25 - (3 + (12 \div 4)) \times 3$ 1

(a) 31 **(b)** 5 **(c)** 7 **(d)** 9

19. 3.45 m expressed in cm is: 1

(a) 3450 **(b)** 345 **(c)** 34.5 **(d)** 34500

20. If $m = 4$, the value of $m + 3m$ is: 1

(a) 17 **(b)** 16 **(c)** 18 **(d)** 15

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Convert 23,45,678 from the Indian to the international system. 2

22. Simplify $8 + (12 - 4) \times 3$. 2

23. Find 3.2×0.5. 2

24. The number of matchsticks needed for a row of n squares is $3n + 1$. How many matchsticks are needed for 5 squares? 2

25. Two supplementary angles are such that one is 2 times the other. Find them. 2

26. Show that $(10a + b) + (10b + a) = 11(a + b)$ and verify for $a = 3$, $b = 5$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Write 34,56,789 in expanded form using powers of 10 and say how many lakhs it is (approx. to one decimal place). 3

28. Evaluate $50 - (4 + (12 - 3) \times 2)$. 3

29. Cloth costs ₹125.5 per metre. Find the cost of 3.5 m of cloth. 3

30. In a pattern, figure n has $3n + 2$ dots. Find the number of dots in figure 4 and in figure 5. How many more dots does figure 5 have? 3

31. Three of the four angles around a point are 95°, 85° and 110°. Find the fourth angle and name the kind of each of the four angles. 3

OR

Apply Kaprekar's routine twice to 5296. 3

32. In isosceles $\triangle ABC$ with $AB = AC$, $\angle A = (x + 10)^\circ$ and $\angle B = (2x - 5)^\circ$. Find x and all three angles. 3

OR

A rectangle has sides $\frac{3}{4}$ m and $\frac{2}{5}$ m. Show its area as the product of the sides and compare it with the area of the smaller side squared. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. A government school uses 1,250 litres of water per day. Find the water used in a year of 365 days, round it to the nearest lakh and write the rounded value in words. 5

34. A fruit seller buys 20 kg of apples at ₹80 per kg and 15 kg of mangoes at ₹120 per kg. He sells all the fruit for ₹3700. Write an expression for his profit or loss and find it. 5

35. A shopkeeper buys 2.5 kg of rice at ₹48 per kg and 1.5 kg of wheat at ₹36 per kg. Find the total cost. He pays ₹200; find the balance. 5

36. A school buys n notebooks at ₹12 each and 4 pens at ₹5 each. Write an expression for the total cost. Find the cost when $n = 10$ and the change from ₹200. 5

OR

A transversal cuts two parallel lines and one of the eight angles is $50°$. How many of the eight angles measure $50°$ and how many measure $130°$? Give reasons. 5

37. For the three-digit number 732, find its reverse, the difference, and show that the difference is divisible by 99. 5

OR

The angles of a triangle are $(x + 10)°$, $(x - 10)°$ and $(2x + 20)°$. Find x, the three angles and classify the triangle. Also find the three exterior angles (one at each vertex). 5

38. A journey of 120 km is done in three parts. The first part is $\frac{1}{4}$ of the journey; $\frac{2}{3}$ of the remaining distance is covered next. Find both distances and the distance still left. 5

OR

The population of a state is 2,50,00,000. A survey covers 4 of every 100 persons. Find the number of people surveyed. 5

END OF QUESTION PAPER Total: 80 Marks

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